

Mesoamerican nephropathy: pathology in search of etiology



Mark Haas¹

Mesoamerican nephropathy is a progressive, often fatal form of tubulointerstitial nephritis affecting young agricultural laborers in Central America. Initially described as a chronic disease, a study by Fischer and coworkers in this issue of *Kidney International* suggests that Mesoamerican nephropathy goes through an active, inflammatory phase. Although the pathologic findings are nonspecific and the etiology of Mesoamerican nephropathy remains unclear, inflammatory infiltrates in areas of evolving and established interstitial fibrosis appear to cause progressive kidney injury.

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see clinical investigation on page 681

Mesoamerican nephropathy (MeN) is a progressive form of tubulointerstitial nephritis that primarily affects young agricultural laborers in Central America, often leading to end-stage renal disease and death.¹ The first case series of MeN with renal biopsy data was published in 2013, although individual cases date back to the 1970s.² To date, at least 20,000 people have died as a result of MeN.¹ Furthermore, similar lesions have been noted in agricultural workers in poor, tropical countries in other parts of the world, including Sri Lanka, leading to the alternative term chronic interstitial nephritis in agricultural communities (CINAC).³ The etiology or etiologies of MeN/CINAC are not known. Early pathologic descriptions of the lesion noted interstitial fibrosis and tubular atrophy (IFTA), with prominent accompanying glomerulosclerosis and variable but not severe interstitial inflammation,² raising repeated episodes of dehydration with associated ischemic injury and/or direct tubular toxicity due to agricultural product(s) such as pesticides as potential

etiologies. In this issue of *Kidney International*, Fischer and coworkers⁴ (2018) report a pathologic study of patients biopsied during an earlier (although still relatively advanced, with creatinine clearances between 32 and 62 ml/min) phase of MeN, characterized by renal insufficiency that was persistent after administration of i.v. fluids, with leukocytosis, neutrophilia, and leukocyturia in the absence of a positive blood culture, and no history of hypertension, diabetes, or heart disease. The renal biopsies in the series from Fischer *et al.*⁴ differed from those in previous series in that there was generally patchy but quite prominent interstitial inflammation in addition to and often associated with areas of IFTA, involving the cortex and frequently the region of the corticomedullary junction. This inflammation consisted mainly of T lymphocytes, although multiple cases showed focal intratubular neutrophils and cellular debris. Notably, there was relative sparing of glomeruli and arteries, absence of intratubular crystals, and no evidence of an immune complex-mediated etiology.⁴

The focal intratubular neutrophils, together with the leukocytosis and neutrophilia seen in many of the patients, suggest the possibility of an infectious etiology for MeN, and Fischer *et al.*⁴ appear to favor such an etiology while not

ruling out a toxic or drug-induced etiology. However, if this were the case, the infection would likely have to be of a kind other than a typical ascending bacterial pyelonephritis; it is worth noting that only one of the authors' 11 patients had frequent urinary tract infections, all but two had no or scarce bacteria on urinalysis, and the biopsies showed sparing of the deeper medulla. In addition, the absence of eosinophilia in all patients tends to argue against a parasitic infection.

Although an infectious etiology of MeN cannot be excluded, the histology and immunohistology of the cases reported by Fischer *et al.*⁴ actually draw a number of important parallels to chronic, active cell-mediated tubulointerstitial nephritis of the types reported in T cell-mediated renal allograft rejection (TCMR) and some examples of chronic, drug-induced interstitial nephritis. The progressive renal injury in these cases results from, or appears to be the result of cell-mediated immunity. In tubulointerstitial lesions of TCMR, the initial pathologic finding is interstitial inflammation in the cortex composed mainly of T lymphocytes with accompanying interstitial edema and lymphocytic tubulitis. If not effectively treated, this is followed by the development of IFTA with inflammation persistent within scarred areas of the cortex, the latter now often referred to in the transplant community as i-IFTA.⁵ Although once thought to represent nonspecific inflammation and thus omitted as a form of rejection-associated injury in the original Banff classification for renal allograft pathology, i-IFTA (Figure 1, left panel) has clearly been shown to be associated with graft failure and loss, to a much greater extent than bland, uninfamed IFTA (Figure 1, right panel).⁶ Furthermore, recent studies have demonstrated that i-IFTA in the renal allograft represents a sequela of active TCMR, often associated with inadequate or suboptimal immunosuppression.⁵

This hypothesis that the pathogenesis of MeN (or CINAC) involves persistent T cell-mediated immune injury progressing from an active state resembling acute TCMR or drug-induced interstitial nephritis, to a chronic, active state characterized by

¹Department of Pathology and Laboratory Medicine, Cedars-Sinai Medical Center, Los Angeles, California, USA

Correspondence: Mark Haas, Department of Pathology and Laboratory Medicine, Cedars-Sinai Medical Center, Los Angeles, California 90048, USA. E-mail: mark.haas@cshs.org

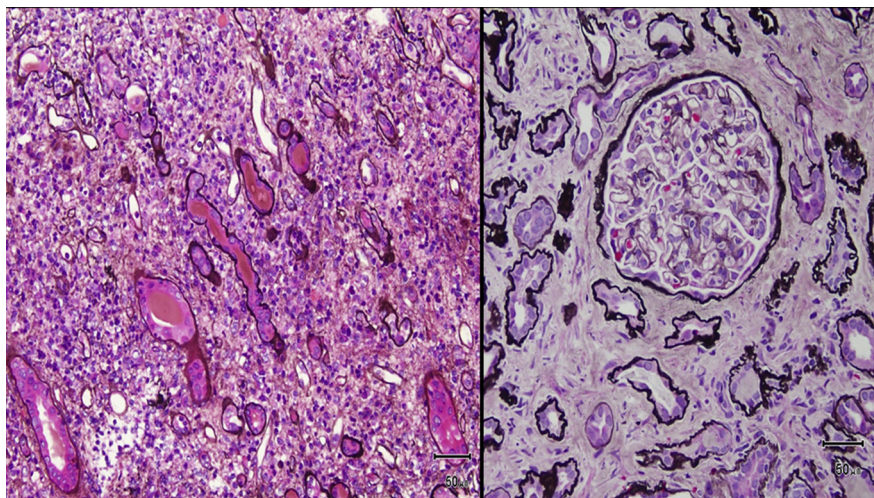


Figure 1 | Histologic comparison of chronic, active tubulointerstitial nephritis (left panel) and inactive tubular atrophy and interstitial fibrosis (bland interstitial fibrosis and tubular atrophy; right panel). The former is seen with chronic, active inflammatory lesions of Mesoamerican nephropathy as shown by Fischer *et al.*,⁴ and resembles i-IFTA seen in renal allografts with persistent T-cell-mediated rejection that has been shown to be associated with high rates of graft loss. Note the presence of interstitial edema in addition to fibrosis, as well as lymphocytic tubulitis in the lower left quadrant of the photomicrograph. By contrast, the lesion in the right panel would be consistent with a late, inactive stage of Mesoamerican nephropathy that is unlikely to be responsive to treatment. Both photomicrographs: Jones methenamine silver stain, original magnification $\times 200$, scale bar 50 μm . To optimize viewing of this image, please see the online version of this article at www.kidney-international.org.

i-IFTA and resembling the cases reported by Fischer *et al.*,⁴ to a scarred and relatively inactive state resembling that in earlier descriptions of MeN,² while intriguing, raises a number of key questions. First, what are the inciting agents/events? Second, why are agricultural workers in Central America and other regions of the world particularly susceptible to this lesion? Third, although the histology of MeN is nonspecific, is there a way using molecular diagnostics to establish that MeN involves similar mechanisms to TCMR? Finally, can the information from these three questions be used to prevent and/or treat MeN?

As Fischer *et al.*⁴ note, identification of inciting agents and/or events will require working closely with the physicians and other health professionals treating patients with MeN/CINAC in regions where this lesion is prevalent. Are there specific compounds within or applied to the crops (typically sugarcane) harvested by the patients? Do the workers ingest or otherwise expose themselves to drugs or other substances that can, in susceptible individuals, trigger an immune response within the renal parenchyma? Next, are

there genetic features of agricultural workers in regions where MeN/CINAC is endemic that differ from those of populations doing similar work in other parts of the world, and predispose the former to the disease? A notable parallel here might be the APOL1 gene; risk alleles of this gene, mainly in individuals of African descent, are associated with the development of focal segmental glomerulosclerosis, including collapsing glomerulopathy in patients infected with human immunodeficiency virus.⁷ Although APOL1 risk alleles confer susceptibility to focal segmental glomerulosclerosis, most individuals with two such alleles do not develop the disease,⁷ suggesting other factors are required as well.

With regard to determining a potential role of T cell-mediated immunity in the pathogenesis of MeN, advances in molecular pathology developed for analysis of renal transplant biopsies may be helpful. Over the past decade, Halloran and colleagues have studied the expression of multiple pathogenesis-based transcripts sets in tissue from a large number of renal allograft biopsies with TCMR, antibody-mediated rejection, and a number of different nonrejection lesions. From these

studies, they have developed a “molecular TCMR score” based on expression of 30 nonredundant transcripts; with high specificity (95%) and accuracy (89%), this molecular score distinguishes renal allograft biopsies showing TCMR from those showing non-TCMR lesions, including antibody-mediated rejection and BK virus nephropathy.⁸ Using biopsies with MeN in an active phase, it appears feasible that a similar approach could be taken to identify pathogenesis-based transcripts associated with active MeN and compare these with pathogenesis-based transcripts seen with TCMR and other inflammatory lesions in transplant and native kidneys to get a better understanding of the pathogenic mechanisms underlying MeN. With this information, potential therapeutic approaches might be developed to reverse or slow the progression of MeN if diagnosed when there is still sufficiently preserved renal parenchyma. In addition, should there be pathogenic similarities between MeN and TCMR, new therapeutic approaches developed for the latter could also be helpful in treating the former. Encouraging evidence for the likelihood of such similarities comes from a preliminary study reporting considerable overlap between genes upregulated in renal transplant biopsies with i-IFTA and native renal biopsies with interstitial nephritis.⁹

In summary, the study of Fischer *et al.*⁴ is potentially important because it identifies MeN/CINAC in an inflammatory stage, possibly amenable to treatment, as opposed to a lesion characterized only by advanced scarring. With this knowledge, it is hoped that work toward understanding the pathogenesis of this condition, including identification of possible etiologic agents and/or initiating events, potential genetic factors that render agricultural workers in Central America and other endemic areas particularly susceptible to it, and molecular characterization of gene expression patterns in tissue from renal biopsies with active MeN, can now proceed in the hope of preventing and/or slowing the progression of the devastating renal injury seen in affected patients.

DISCLOSURE

The author declared no competing interests.

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Total kidney volume: the most valuable predictor of autosomal dominant polycystic kidney disease progression



Cheng Xue¹, Chenchen Zhou¹ and Changlin Mei¹

Autosomal dominant polycystic kidney disease (ADPKD) is characterized by the progressive development of kidney cysts resulting in damage to kidney structure and function. Yu *et al.* showed that height-adjusted total kidney volume (htTKV) was a significant independent predictor of ADPKD progression after a follow-up of 14.5 years. The genotype was not an independent prognostic factor after adjusting for htTKV. This commentary discusses several considerations in model comparison, biomarker validation, and future challenges in ADPKD research.

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see clinical investigation on page 691

Autosomal dominant polycystic kidney disease (ADPKD) is the most common hereditary renal disease, accounting for 5% to 10% of the causes of end-stage renal disease

(ESRD) in adults worldwide.¹ ADPKD is genetically heterogeneous, with 2 main causative mutations: *PKD1* (16p13.3, 80% of cases) and *PKD2* (4q21, 15% of cases).¹ Mutations lead to the formation and development of fluid-filled cysts that cause progressive loss of kidney function. The natural course of ADPKD varies significantly between and within affected families, with the age of reaching ESRD being from childhood to older than 80 years of age.¹ Drugs that inhibit cyst growth

always show the greatest benefit in the rapid ADPKD progression stage but do not show significant improvements in early and later stages.² Therefore, the identification of predictors of ADPKD progression has become increasingly important for predicting prognosis and screening the most possible beneficiaries of early interventions in clinical practice.

In recent decades, several important predicting factors associated with the progression and ESRD risk of ADPKD have been investigated.^{2–5} These factors include total kidney volume (TKV), genotype (*PKD1*, particularly the truncating mutation), male sex, the early onset of hypertension, overt proteinuria, macroalbuminuria, cyst hemorrhages, and cyst infections.⁵ Among these factors, TKV has been qualified by the US Food and Drug Administration as a prognostic biomarker in ADPKD and is widely used as an endpoint in most clinical trials.² Genotype is an equally important biomarker of progression, as shown by 2 important cohort studies.^{3,4}

TKV as an independent predictor of ADPKD progression

In this issue, Yu *et al.*⁶ showed that baseline height-adjusted TKV (htTKV) was a significant independent predictor of the risk of chronic kidney disease (CKD) progression in ADPKD patients after a follow-up of 14.5 years. The results are solid, with a lower dropout rate. The risks of reaching each CKD stage per 100 ml/m increment of htTKV were 1.38 for CKD stage 3, 1.42 for stage 4, 1.35 for ESRD, 1.61 for glomerular filtration rate (GFR) decreased $\geq 30\%$, and 1.43 for GFR decreased $\geq 57\%$. In the models testing the association of genotype with CKD progression, Yu *et al.*⁶ found that the genotype was not an independent prognostic factor after adjusting for htTKV. Furthermore, Yu *et al.*⁶ compared the goodness-of-fit and prognostic ability of models between genotype and htTKV; the model did not change significantly by adding genotype to the htTKV-based model. However, adding htTKV to the genotype-based model improved the significance and

¹Kidney Institute, Department of Nephrology, Changzheng Hospital, Second Military Medical University, Shanghai, China

Correspondence: Changlin Mei, Kidney Institute, Department of Nephrology, Shanghai Changzheng Hospital, Second Military Medical University, Shanghai 200003, China. E-mail: chlmei1954@126.com